

Analog IC Design (Xschem, Ngspice, ADT)**Lab 07****Five-Transistor OTA Design****Intended Learning Objectives**

In this lab you will:

- Design and simulate a 5T OTA.
- Learn how to plot and use gm/ID design charts.
- Learn how to simulate the open-loop characteristics of the 5T OTA.
- Learn how to simulate the closed-loop characteristics of the 5T OTA.

Part 1: OTA Design

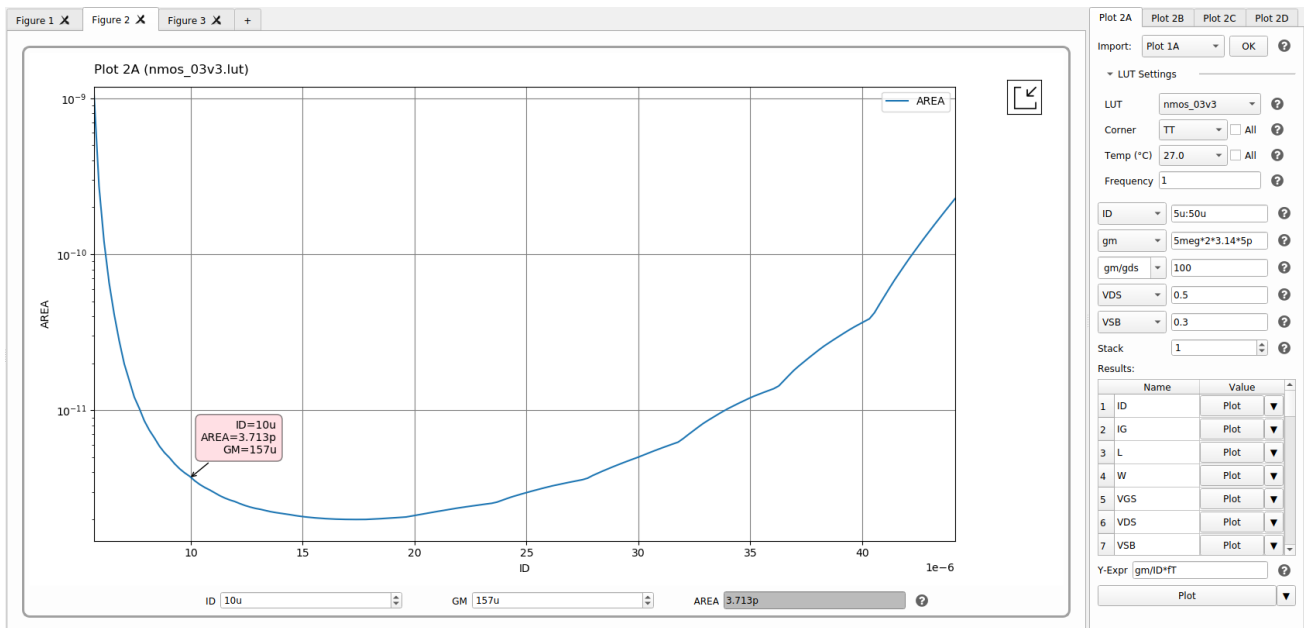
Use gm/ID methodology to design a diff input SE output operational transconductance amplifier (OTA) that achieves the following specs. Use an ideal external ideal current source in your test bench and design your own current mirror.

Parameter	Value
Supply voltage	1.8V
Load	5pF
Open loop DC voltage gain	$\geq 34\text{dB}$
CMRR @ DC	$\geq 74\text{dB}$
Phase margin (PM)	$\geq 70^\circ$
Current seed (reference current)	10uA
CM input range – low	$\leq 1.2\text{V}$
CM input range – high	$\geq 1.5\text{V}$
GBW	$\geq 5\text{MHz}$
OTA current consumption	Minimize
Area	Minimize

Design Considerations:

- 1) Since the required gain is not very high, a simple single-stage amplifier will be sufficient.
- 2) From the CMIR spec you need to identify the input pair type: NMOS or PMOS?
 ➔ **Hint:** Is the CMIR closer to the GND rail or the VDD rail?
- 3) In order to determine the OTA current consumption two approaches can be used: a simple approach and an optimized approach. Both of them are acceptable.
 - First, the simple approach:
 - i. Assume a reasonable gm/ID for the input pair, e.g., gm/ID=15.
 - ii. Use the GBW spec to calculate the required bias current.

- iii. The mirroring ratio (from the current seed) needs to be reasonable because a current mirror must be practically implemented using unit elements. Round the current to a reasonable value.
 - iv. Recalculate the g_m/I_D given the rounded current.
 - Second, the optimized approach:
 - i. Sweep the current while using g_m as knob and g_m/g_{ds} as condition
 - ii. Plot AREA vs I_D (note that the OTA tail current will be double the I_D)
 - iii. From the graph it is clear that picking $I_D > 15\mu A$ doesn't make sense because you will be losing both AREA and power.
 - iv. Selecting $I_D = 10\mu A$ (less power more area) or $I_D = 15\mu A$ (more power less area) both seem reasonable options and will give reasonable mirroring ratios.
 - v. However, given $V_{incm-min}$, we need to pick $I_D = 10\mu A$ to which corresponds to higher g_m/I_D and lower VGS.
- ➔ Hint: Assume the tail current source has $V_{DS} = 0.3$, and plot $V_{incm} = V_{GS} + 0.3$ for the input pair.



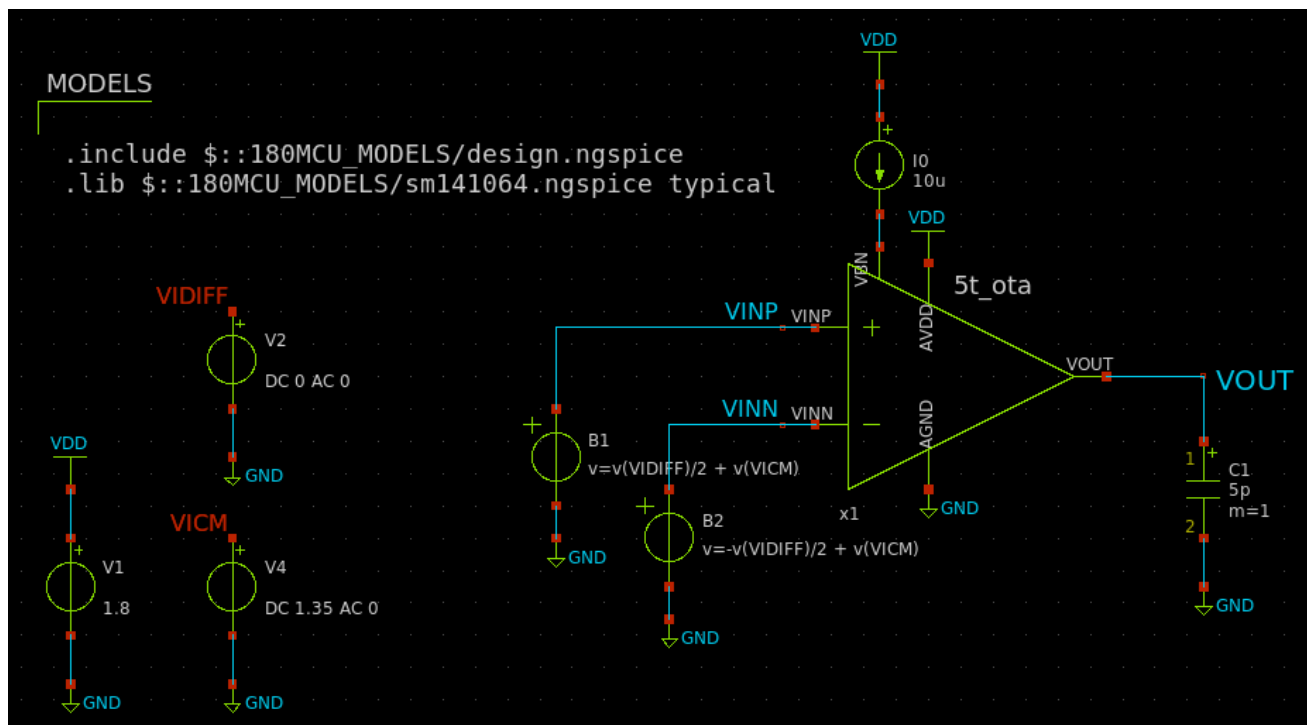
- 4) Since this is a single stage OTA, the PM spec will be probably satisfied if reasonable sizing is selected (do not use excessively large W). No extra design effort is needed.
- 5) For the tail current source:
 - CMRR is g_m/g_{ds} higher than A_{vdc} . $CMRR = A_{vdc} + 40dB$ means $2 \cdot g_{m,load} \cdot R_{ss} = 100$ which can be satisfied with a simple tail current source ($R_{ss} = r_o$).
 - Even if it is not explicitly mentioned, you should always make sure that λ makes sense, e.g. < 0.1 . Note that λ depends on V_{DS} , so it is good to check it at the worst case ($V_{DS} = 0.3$).
 - Even if it is not explicitly mentioned, you should always make sure that the mismatch makes sense, e.g., 1-2% standard deviation.

Report the following:

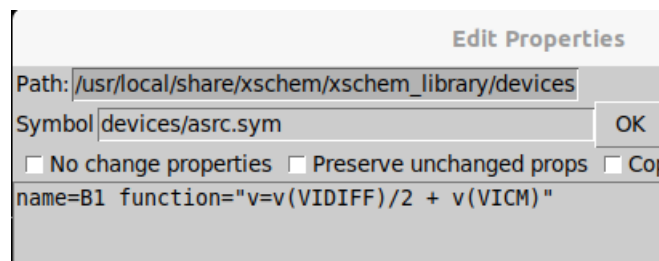
- 1) Detailed design procedure and hand analysis. You need to explain why you chose the architecture that you implemented.
- 2) A table showing W , L , g_m , I_D , g_m/I_D , V_{dsat} , $V_{ov} = V_{GS} - V_{TH}$, and $V^* = \frac{2}{g_m/I_D}$ of all transistors (as calculated from the Sizing Assistant).

Part 2: Open-Loop OTA Simulation¹

Create a testbench as shown below.



→ Hint: Use asrc at the input and output instead of the baluns to combine and separate common-mode and differential signals.



→ Note that IDC connection (sinking or sourcing) in the test bench may be different from the one shown above depending on the type of your input pair (PMOS/NMOS).

Report the following:

- 1) Schematic of the OTA with DC node voltages clearly annotated.
 - Use VICM at the middle of the CMIR.
 - Is the current (and gm) in the input pair **exactly** equal?
 - What is DC voltage at VOUT? Why?
- 2) Diff small signal ccs:
 - Use AC analysis (1Hz:10GHz, logarithmic, 10 points/decade).

¹ An OTA can be simulated without closing the loop if it is perfectly matched (zero offset voltage). For this 5T OTA in open loop, the output node DC level will follow the diode connected node. For two-stage OTA, the systematic offset voltage may drive the output to one of the rails. Also, when simulating mismatch (Monte Carlo simulation), open-loop simulation cannot be used. You must close the loop in the cases of random and systematic offset, so that the offset voltage is automatically adjusted by the feedback action and the dc bias is correctly set.

- Set VID AC = 1 and VICM AC = 0.
 - Use VICM DC at the middle of the CMIR.
 - Plot diff gain (in dB) vs frequency.
 - Compare simulation results with hand calculations in a table.
- **Hint: In the waveform graph, you can plot db20 and unwrapped phased using the expressions below:**

```
Av DIFF dB; vod db20()
Av DIFF Phase; ph(vod) cph()
```

3) CM small signal ccs:

- Use AC analysis (1Hz:10GHz, logarithmic, 10 points/decade).
- Set VICM AC = 1 and VID AC = 0.
- Use VICM DC at the middle of the CMIR.
- Plot CM gain in dB vs frequency.
- Compare simulation results with hand calculations in a table.

4) CMRR:

- Use VICM DC at the middle of the CMIR.
- Plot CMRR in dB vs frequency.
- Compare simulation results with hand calculations in a table.

→ Example ngspice code for parts 1 to 4:

```
.control

alterparam sw_stat_global = 0
alterparam sw_stat_mismatch = 0
reset

save all

let x = 1
set num = {$&x}
dowhile x <= 6
save @m.x1.xm{$num}.m0[id]
save @m.x1.xm{$num}.m0[vgs]
save @m.x1.xm{$num}.m0[vds]
save @m.x1.xm{$num}.m0[vdsat]
save @m.x1.xm{$num}.m0[vth]
save @m.x1.xm{$num}.m0[gm]
save @m.x1.xm{$num}.m0[gds]
save @m.x1.xm{$num}.m0[gmb]
save @m.x1.xm{$num}.m0[cgg]
save @m.x1.xm{$num}.m0[cdd]
save @m.x1.xm{$num}.m0[cgs]
let x = x + 1
set num = {$&x}
end
```

```

* === OP Point ===
op
write 5t_ota_tb_ol.raw
set appendwrite

echo \"          === MOSFET Operating Points ===\"
show m : id : vgs : vds : vth : vdsat : gm : gds : gmbs

echo \"          === Bias Points ===\"
let x = 1
dowhile x <= 6
    set num = {$&x}
    let gmid = @m.x1.xm{$num}.m0[gm]/@m.x1.xm{$num}.m0[id]
    let Vstar = 2/gmid
    echo \"M{$num}: gm/id = $&gmid S/A, Vstar = $&Vstar V\"
    let x = x + 1
end

* === DIFFERENTIAL AC ===
alter V2 AC = 1
alter V4 AC = 0
ac dec 10 1 10e9

* === COMMON MODE AC ===
alter V2 AC = 0
alter V4 AC = 1
ac dec 10 1 10e9

* === EXTRACT RESULTS ===
set units=degrees
let vod = ac1.v(VOUT)
let vocm = ac2.v(VOUT)
let CMRR = vdb(vod) - vdb(vocm)

* === MEASUREMENTS ===
meas ac Av_diff_db find vdb(vod) at=1
meas ac Av_diff_mag find vmag(vod) at=1
let A3db = Av_diff_mag/sqrt(2)
meas ac BW_diff when vmag(vod)=A3db fall=1
meas ac UGF when vmag(vod)=1 fall=1
meas ac P_UGF find vp(vod) at=UGF
let GBW = Av_diff_mag * BW_diff
let PM = 180 + P_UGF

meas ac Av_CM_db find vdb(vocm) at=1
meas ac Av_CM_mag find vmag(vocm) at=1
let A3db = Av_CM_mag/sqrt(2)
meas ac BW_CM when vmag(vocm)=A3db fall=1

let CMRR_db = Av_diff_db - Av_CM_db

```

```

* === DISPLAY RESULTS ===
echo \"          === AC Results ===\"
echo \"Av Diff: $&Av_diff_db dB, BW: $&BW_diff Hz\"
echo \"UGF: $&UGF Hz\"
echo \"GBW: $&GBW Hz\"
echo \"PM: $&PM deg\"
echo \"Av CM: $&Av_CM_db dB, BW: $&BW_CM Hz\"
echo \"CMRR: $&CMRR_db dB\"

write 5t_ota_tb_01.raw

.endc

```

5) Avcm vs VICM:

- Use parametric sweep (not DC sweep) for VICM = CMIR-low:10m:CMIR-high.
- Plot CM gain at 1Hz in dB vs VICM.
- Justify the results. Does Avcm increase or decrease vs VICM? Why?

➔ Example ngspice code:

```

.control

alterparam sw_stat_global = 0
alterparam sw_stat_mismatch = 0
reset

save all

let index = 0
compose VICM_vec start=1.2 stop=1.5 step=50m
let GBW_vec = VICM_vec
let Avcm_vec = VICM_vec
setscale VICM_vec

foreach VICM_val $&VICM_vec
alter V4 DC=$VICM_val

* === COMMON MODE AC ===
alter V2 AC = 0
alter V4 AC = 1
ac dec 10 1 10e9

* === MEASUREMENTS ===
meas ac Av_CM_mag find vmag(VOUT) at=1
let Avcm_vec[index] = Av_CM_mag
let index = index + 1
end

write 5t_ota_tb_01.raw Avcm_vec

```

```

plot Avcm_vec vs VICM_vec

write 5t_ota_tb_cmir_ac.raw

.endc

```

6) Diff large signal ccs:

- Use VICM at the middle of the CMIR.
- Use DC sweep (**not parametric sweep**) VID = -VDD:1m:VDD. You must use a small step (1mV) because the gain region is very small (steep slope).
- Plot VOUT vs VID.
- From the plot, what is the value of Vout at VID = 0? Why?
- Plot the derivative of VOUT vs VID. Compare the peak with Avd. Comment.

➔ Example ngspice code:

```

.control

alterparam sw_stat_global = 0
alterparam sw_stat_mismatch = 0
reset

save all

* === DIFFERENTIAL LARGE SIGNAL ===
dc V2 -1.8 1.8 10m

* === EXTRACT EXTREME VALUES ===
let Vmax = maximum(VOUT)
let Vmin = minimum(VOUT)

* === DISPLAY RESULTS ===
echo \"Diff Large Signal Analysis Results:\"
echo \"Max Output: $&Vmax V\"
echo \"Min Output: $&Vmin V\"

write 5t_ota_tb_ol.raw

.endc

```

7) CM large signal ccs (GBW vs VICM):

- Use AC analysis (1Hz:10Gz, logarithmic, 10 points/decade).
- Set VIDAC = 1 and VICMAC = 0.
- Use **parametric sweep** (**not DC sweep**) VICM = 0:50m:VDD.
- Plot GBW vs VICM. Plot the results overlaid on the results of the previous method (region parameter).

➔ Hint: Instead of using parametric sweep, another alternative in this case is to use DC sweep (sweep VICM) and plot gm of the input stage (which will be give the same behavior of the GBW).

- Annotate the CM input range. Calculate the input range as the range over which the GBW is within 90% of the max GBW, i.e., 10% reduction in GBW².

→ **Example ngspice code:**

```
.control

alterparam sw_stat_global = 0
alterparam sw_stat_mismatch = 0
reset

save all

let index = 0
compose VICM_vec start=0 stop=1.8 step=50m
let GBW_vec = VICM_vec
setscale VICM_vec

foreach VICM_val $&VICM_vec
alter V4 DC=$VICM_val

* === DIFFERENTIAL AC ===
alter V2 AC = 1
alter V4 AC = 0
ac dec 10 1 10e9

* === MEASUREMENTS ===
meas ac Av_diff_mag find vmag(VOUT) at=1
let A3db = Av_diff_mag/sqrt(2)
meas ac BW_diff when vmag(VOUT)=A3db fall=1
let GBW = Av_diff_mag * BW_diff

let GBW_vec[index] = GBW
let index = index + 1
end

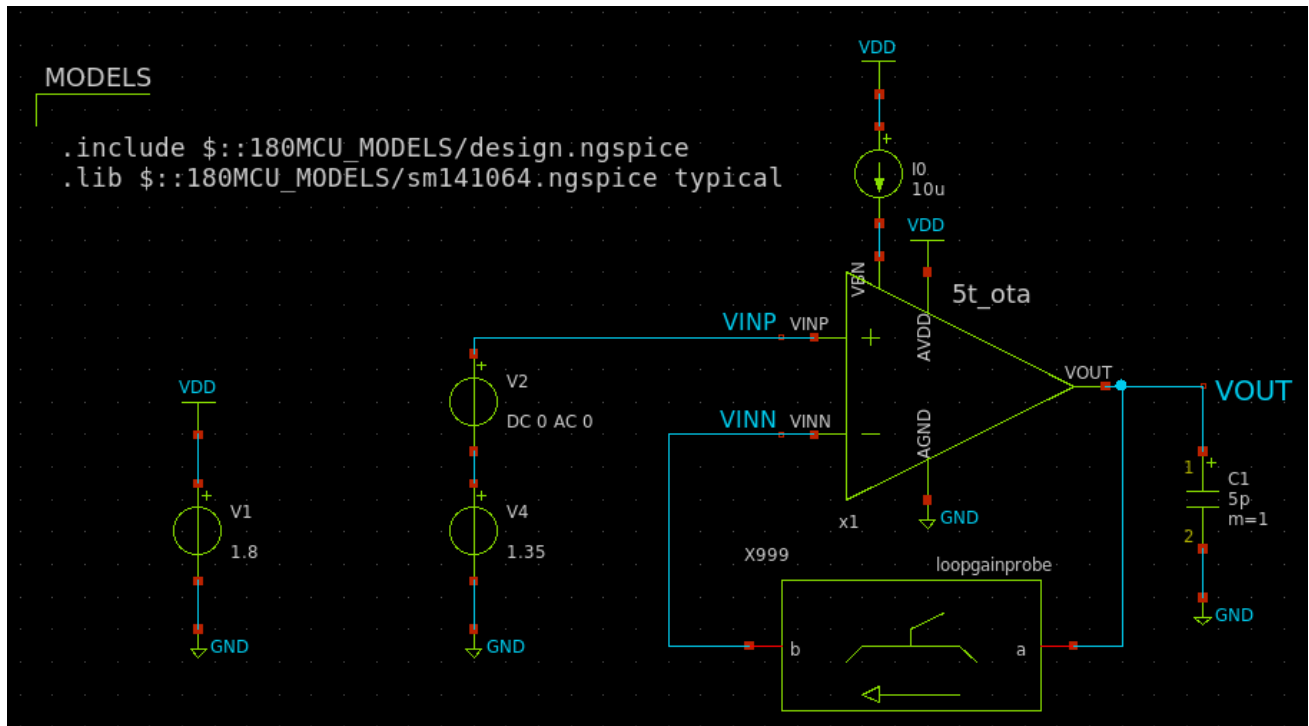
write 5t_ota_tb_01.raw GBW_vec

plot GBW_vec vs VICM_vec

.endc
```

² If you are using NMOS input pair, body effect may cause CMIR to extend till VDD (why?).

Part 3: Closed-Loop OTA Simulation



Create a testbench as shown above. Report the following:

- 1) Schematic of the OTA with DC OP point clearly annotated in unity gain buffer configuration.
 - Set VICM at the middle of the CMIR.
 - Is the current (and gm) in the input pair exactly equal³? Why?
 - Calculate the mismatch in I_D and gm.

→ Example ngspice code:

```
.control

alterparam sw_stat_global = 0
alterparam sw_stat_mismatch = 0
reset

save all

let x = 1
set num = {${x}}
dowhile x <= 6
save @m.x1.xm{${num}}.m0[id]
save @m.x1.xm{${num}}.m0[vgs]
```

³ For the case of the OTA in closed-loop unity-gain buffer connection, the output will follow the input. The output voltage deviates from its CM level (the voltage at the mirror node for the 5T OTA) in order to match the input voltage. Since the gain is finite, there will be a non-zero differential input voltage that will cause an imbalance between the two sides of the differential pair. This imbalance behaves exactly like mismatch. To avoid this imbalance, we may close the loop using Vin equal to the voltage at the mirror node. Any other Vin will introduce mismatch between the gm's of the input pair, and you will have to use Avcm and CMRR equations that consider mismatch.

```

save @m.x1.xm{$num}.m0[vds]
save @m.x1.xm{$num}.m0[vdsat]
save @m.x1.xm{$num}.m0[vth]
save @m.x1.xm{$num}.m0[gm]
save @m.x1.xm{$num}.m0[gds]
save @m.x1.xm{$num}.m0[gmbs]
save @m.x1.xm{$num}.m0[cgg]
save @m.x1.xm{$num}.m0[cdd]
save @m.x1.xm{$num}.m0[css]
let x = x + 1
set num = {$&x}
end

* === OP Point ===
op
write 5t_ota_tb_ol.raw
set appendwrite

echo \"          === MOSFET Operating Points ===\"
show m : id : vgs : vds : vth : vdsat : gm : gds : gmbs

echo \"          === Bias Points ===\"
let x = 1
dowhile x <= 6
    set num = {$&x}
    let gmid = @m.x1.xm{$num}.m0[gm]/@m.x1.xm{$num}.m0[id]
    let Vstar = 2/gmid
    echo \"M{$num}: gm/id = $&gmid S/A, Vstar = $&Vstar V\"
    let x = x + 1
end

write 5t_ota_tb_cl.raw

.endc

```

2) Loop gain:

- Use STB analysis (1Hz:10GHz, logarithmic, 10 points/decade) in unity gain buffer configuration.
- ➔ **The required ngspice code for stb analysis is in the stb.spi file included in the LoopGainProbe folder. You just need to included this file. Change the path as needed. View the file and examine the measure commands.**

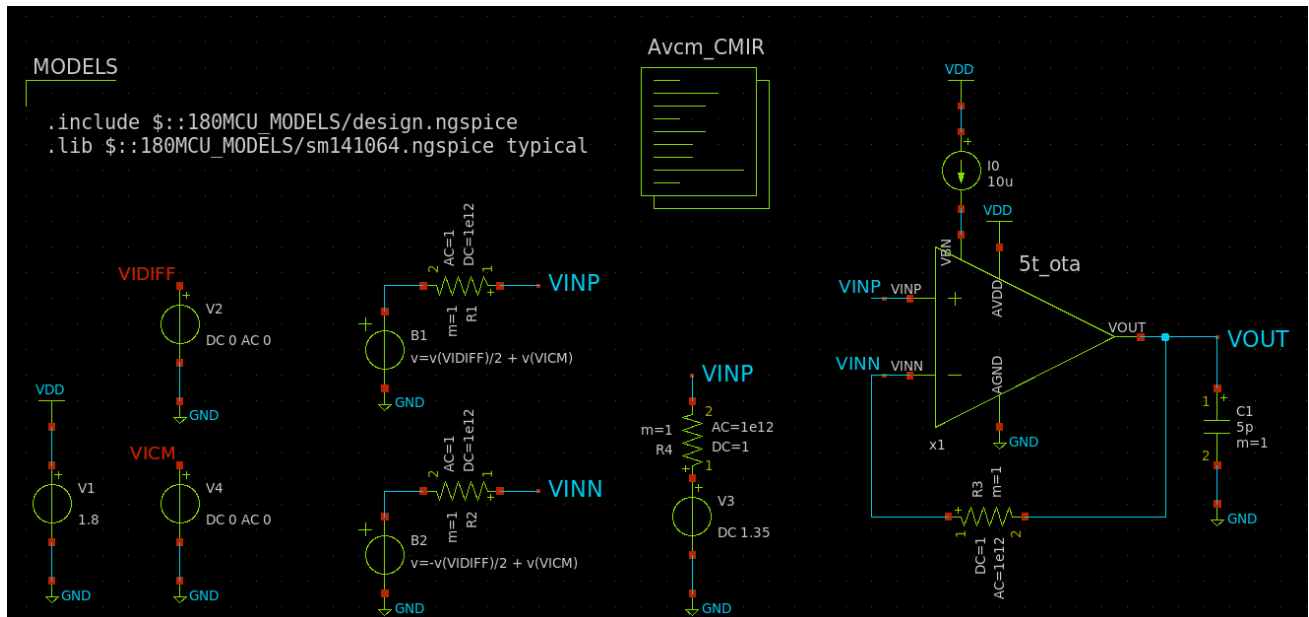
```
.include ~/aic_labs/lab_07/LoopGainProbe/stb.spi
```

- Set VICM at the middle of the CMIR.
- Plot loop gain in dB and phase vs frequency. Use the following expressions in the waveform graph.

```
dB20(LG); tian_signal db20()
Phase(LG) ; ph(tian_signal) cph()
```

- Compare LG DC gain and GX with those obtained from open-loop simulation. Comment.
- Compare simulation results with hand calculations in a table.

Part 4: Effect of Mismatch on CMRR



Copy your testbench in a new cell and modify the new schematic as shown above.

- Resistors with different AC/DC setting (res_ac) are added in order to connect the feedback loop in DC and break it in AC. The DC OP point is set by the unity gain feedback buffer connection, while the AC stimulus is set by the asrc components.
- Note that the DC feedback loop will introduce mismatch in the input pair (why?).

Report the following:

- Use AC analysis (at 1Hz, no frequency sweep). Keep OP simulation enabled because we will plot ID.
- Set VICMAC = 1 and VIDAC = 0.
- Use **parametric** sweep (**not DC sweep**) for VIN = 0.8:50m:1.5.
- Plot ID of the input pair overlaid vs VICM. Comment on the results.
- Plot CM gain at 1Hz vs VIN. Comment on the results.

➔ Example ngspice code:

```
.control

alterparam sw_stat_global = 0
alterparam sw_stat_mismatch = 0
reset

save all

save @m.x1.xm1.m0[id]
save @m.x1.xm2.m0[id]

let index = 0
compose VICM_vec start=0.8 stop=1.5 step=50m
let Avcm_vec = VICM_vec
let id1_vec = VICM_vec
```

```

let id2_vec = VICM_vec
setscale VICM_vec

foreach VICM_val $&VICM_vec
alter V3 DC=$VICM_val

op

let id1_vec[index] = @m.x1.xml.m0[id]
let id2_vec[index] = @m.x1.xm2.m0[id]

* === COMMON MODE AC ===
alter V2 AC = 0
alter V4 AC = 1
ac dec 10 1 10e9

* === MEASUREMENTS ===
meas ac Av_CM_mag find vmag(VOUT) at=1
let Avcm_vec[index] = Av_CM_mag

let index = index + 1
end

write 5t_ota_tb_dc_cl_ac_ol.raw Avcm_vec id1_vec id2_vec

plot Avcm_vec vs VICM_vec
plot id1_vec id2_vec vs VICM_vec

.endc

```

Lab Summary

- In Part 1 you learned:
 - How to design and OTA to meet desired specifications using the gm/ID methodology.
- In Part 2 you learned:
 - How to simulate the small-signal differential gain of a 5T OTA in open-loop configuration.
 - How to simulate the small-signal common-mode gain of a 5T OTA in open-loop configuration.
 - How to simulate the large-signal differential characteristics of a 5T OTA in open-loop configuration.
 - How to simulate the large-signal common-mode characteristics of a 5T OTA in open-loop configuration.
- In Part 3 you learned:
 - How to simulate the loop gain of a 5T OTA in closed-loop configuration.
- In Part 4 you learned:
 - How to simulate the effect of mismatch on the characteristics of a 5T OTA.

Acknowledgements

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